CSC 411: Lecture 02: Linear Regression

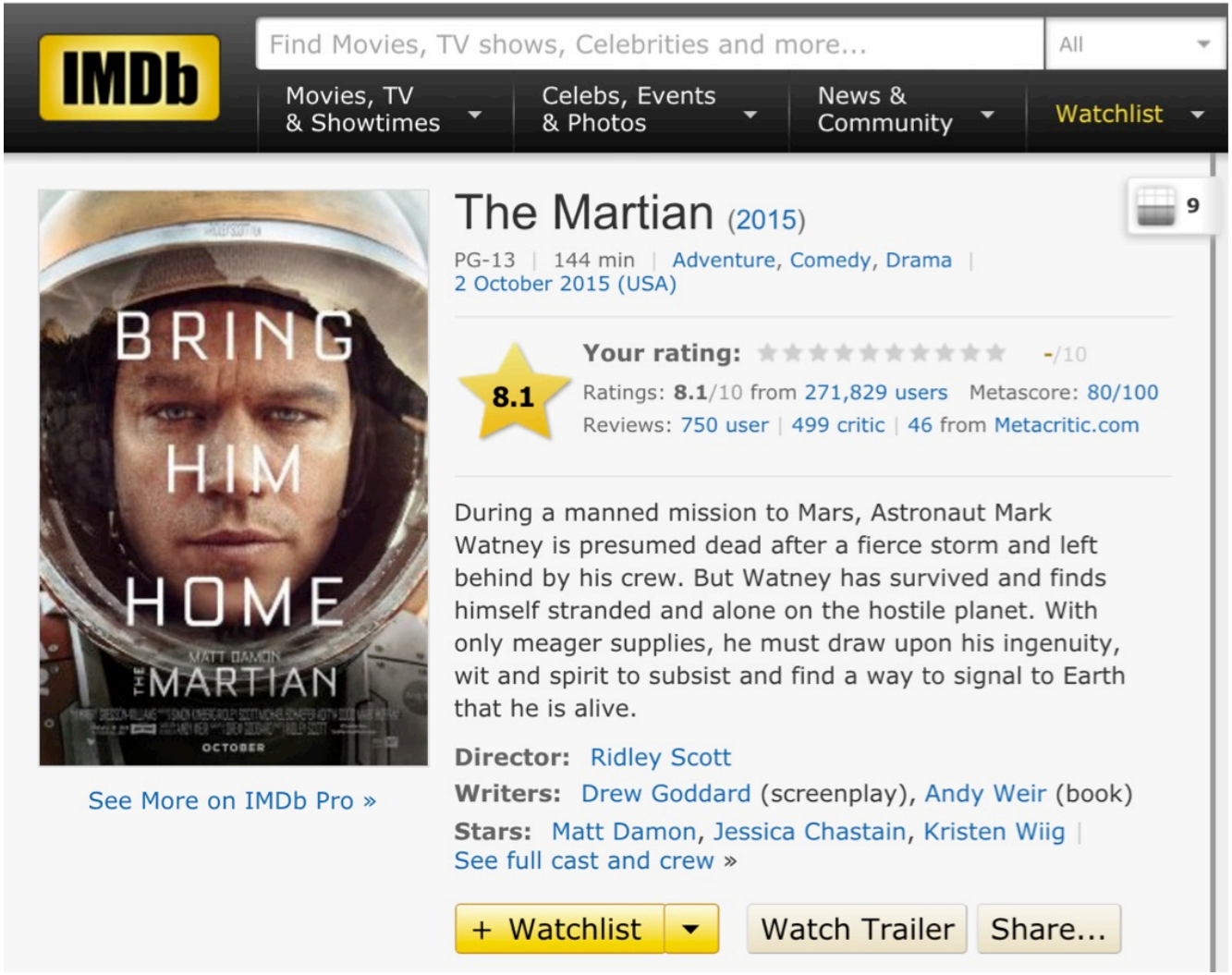
Richard Zemel, Raquel Urtasun and Sanja Fidler

University of Toronto

(Most plots in this lecture are from Bishop's book)

Problems for Today

- What should I watch this Friday?



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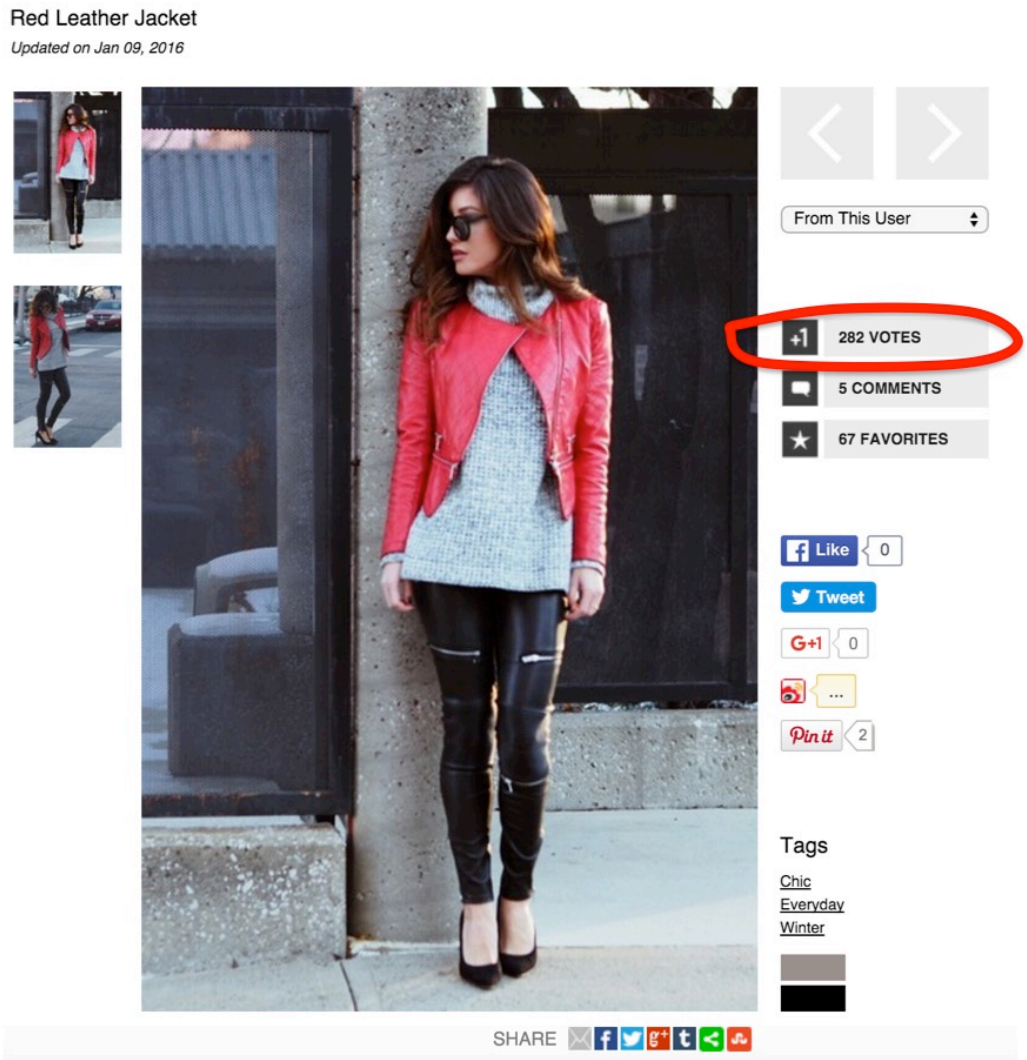
Problems for Today

- **Goal**: Predict movie rating automatically!



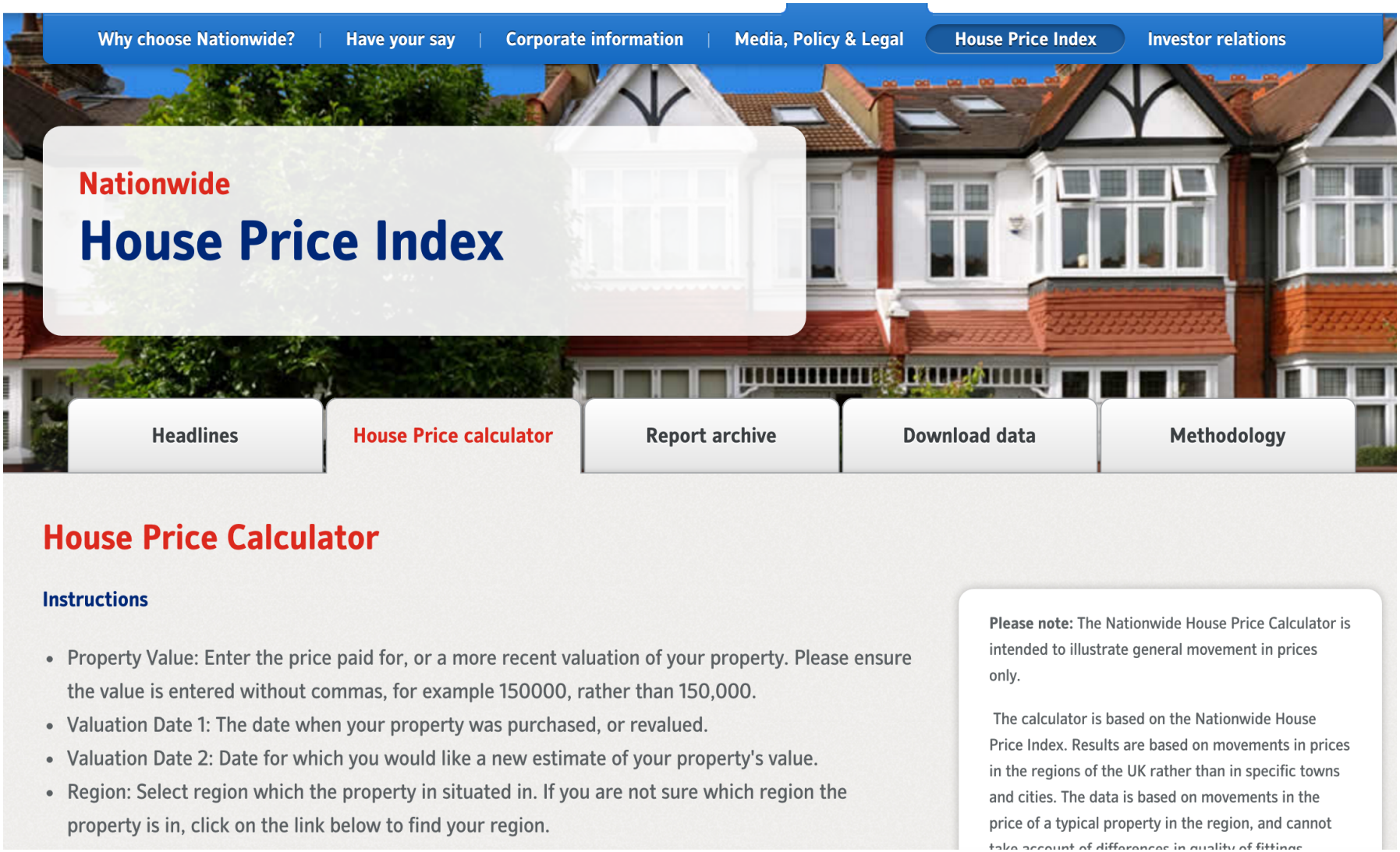
Problems for Today

- **Goal:** How many followers will I get?



Problems for Today

- **Goal:** Predict the price of the house



Regression

- What do all these problems have in common?

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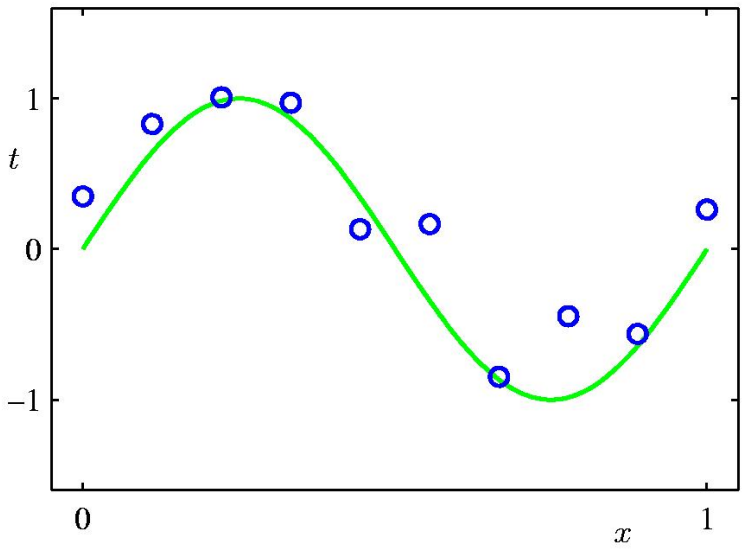
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  - A model, a function that represents the relationship between x and t
  - A loss or a cost or an objective function, which tells us how well our model approximates the training examples
  - Optimization, a way of finding the parameters of our model that minimizes the loss function

Today: Linear Regression

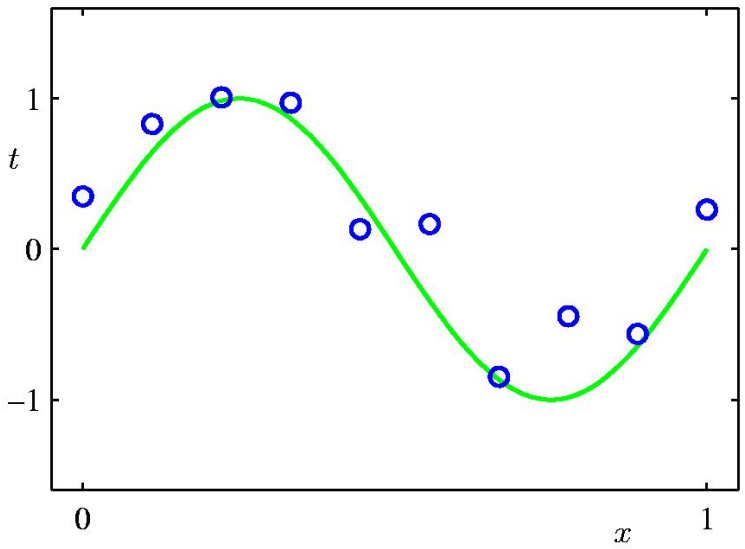
- Linear regression
  - continuous outputs
  - simple model (linear)
- Introduce key concepts:
  - loss functions
  - generalization
  - optimization
  - model complexity
  - regularization

Simple 1-D regression



- Circles are data points (i.e., training examples) that are given to us

Simple 1-D regression

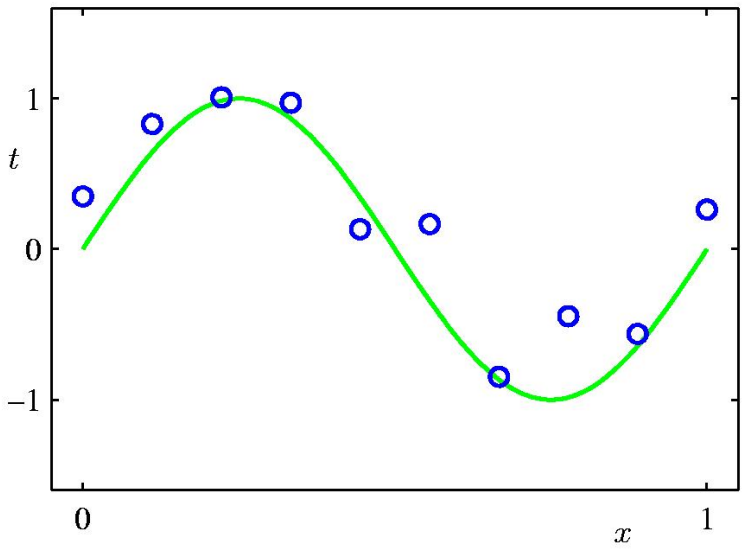


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$$t(x) = f(x) + \epsilon$$

with ϵ some noise

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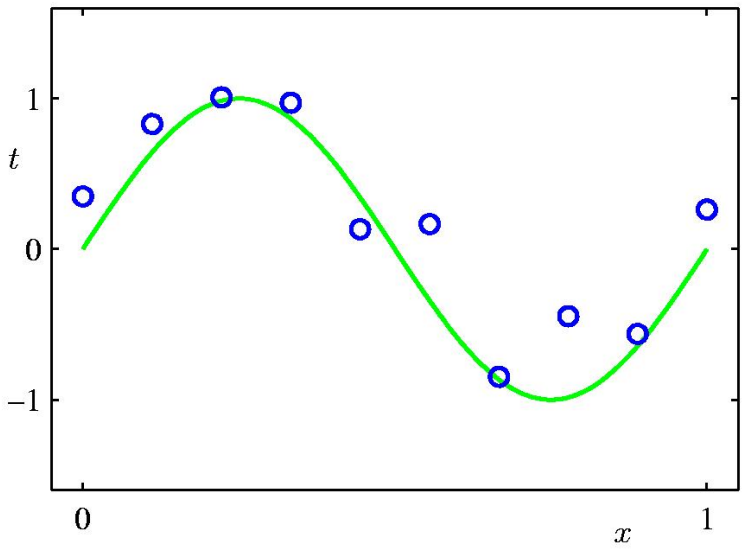
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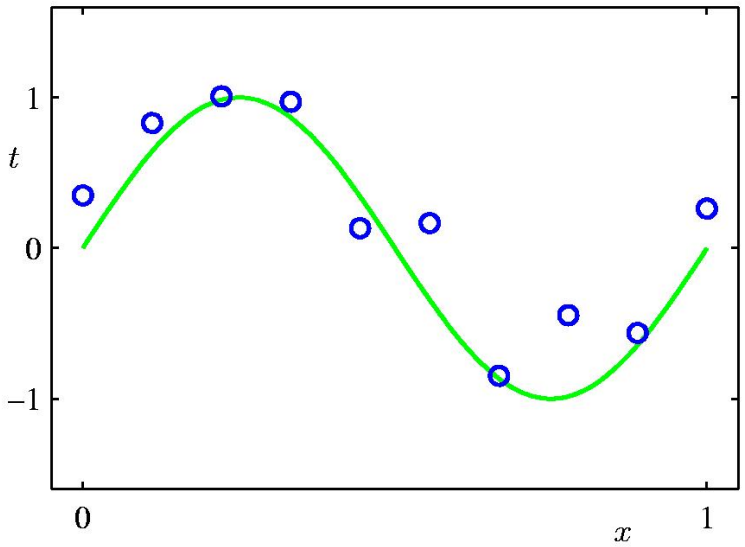
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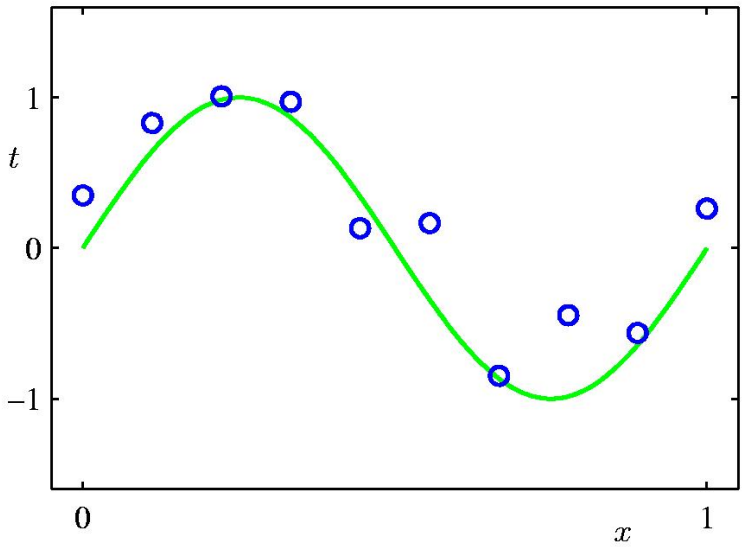
- In green is the "true" curve that we don't know
- Goal: We want to fit a curve to these points

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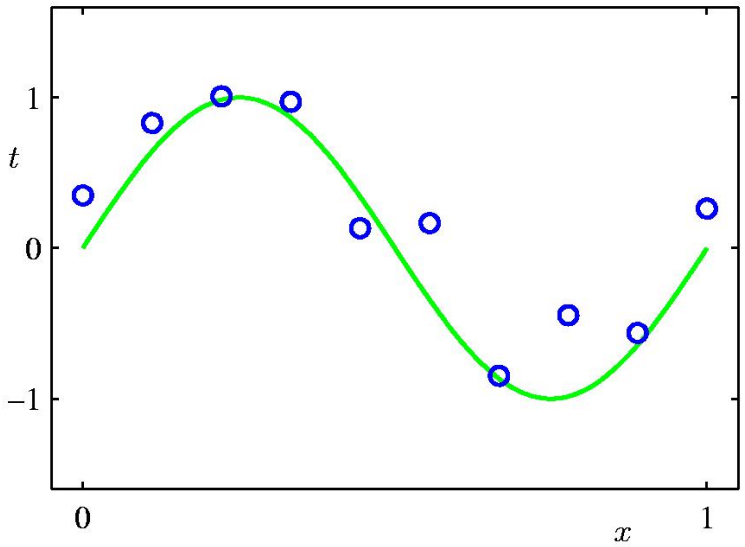
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Simple 1-D regression



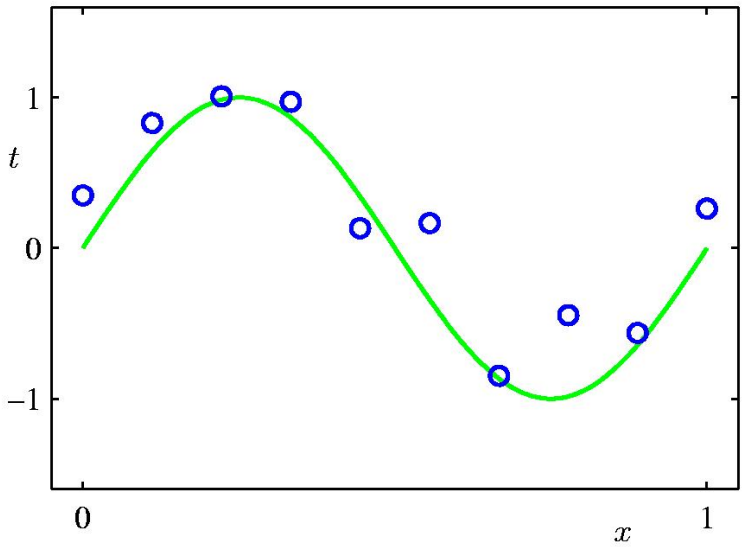
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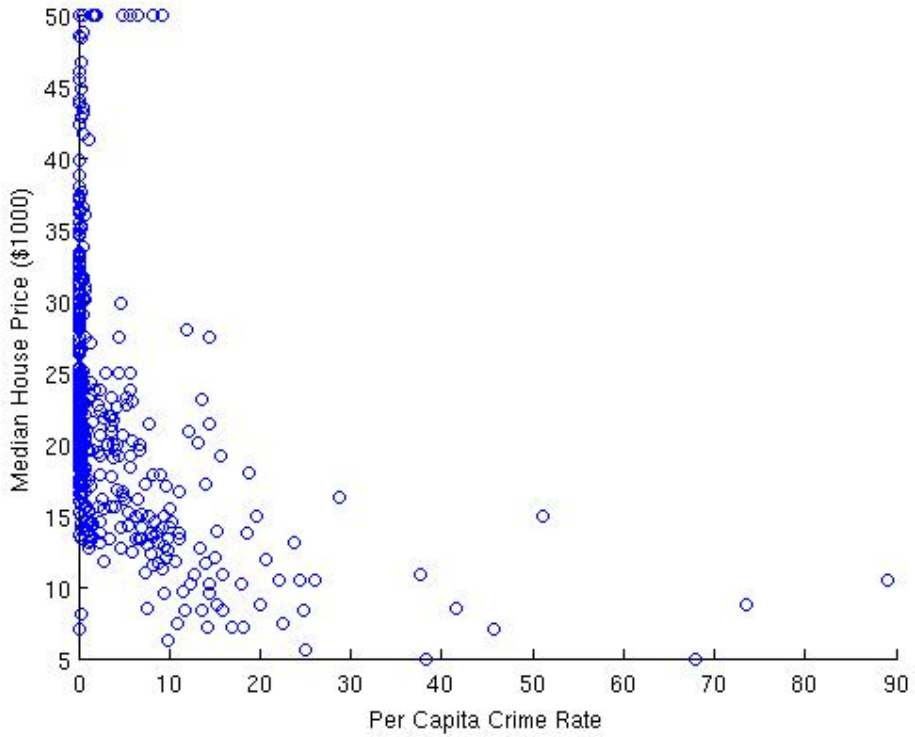
- Key Questions:
  - How do we parametrize the model?
  - What loss (objective) function should we use to judge the fit?
  - How do we optimize fit to unseen test data (generalization)?

Example: Boston Housing data

- Estimate median house price in a neighborhood based on neighborhood statistics

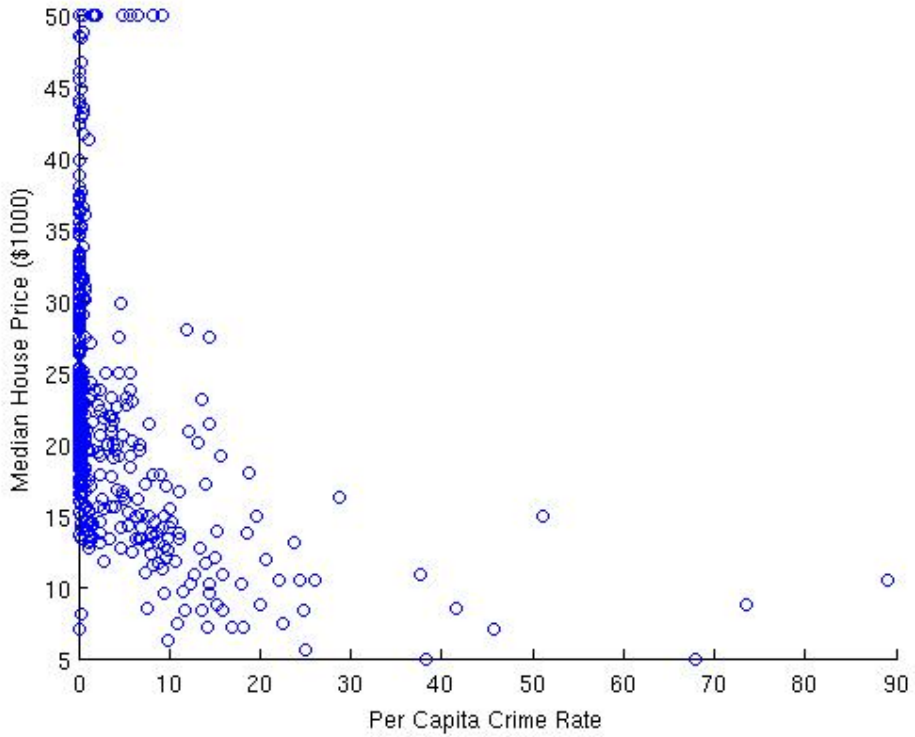
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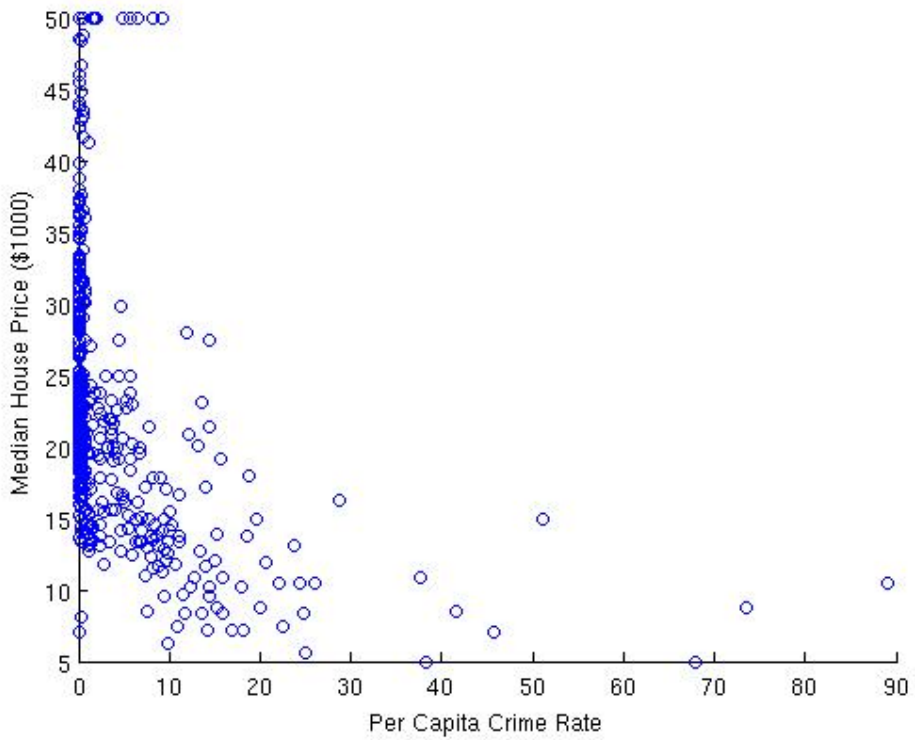
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Example: Boston Housing data

- Estimate median house price in a neighborhood based on neighborhood statistics
- Look at first possible attribute (feature): per capita crime rate



- Use this to predict house prices in other neighborhoods
- Is this a good input (attribute) to predict house prices?

Represent the Data

- Data is described as pairs $\mathcal{D} = \{(x^{(1)}, t^{(1)}), \cdots, (x^{(N)}, t^{(N)})\}$
  - $x \in \mathbb{R}$ is the input feature (per capita crime rate)
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  - Evaluate hypothesis on test set

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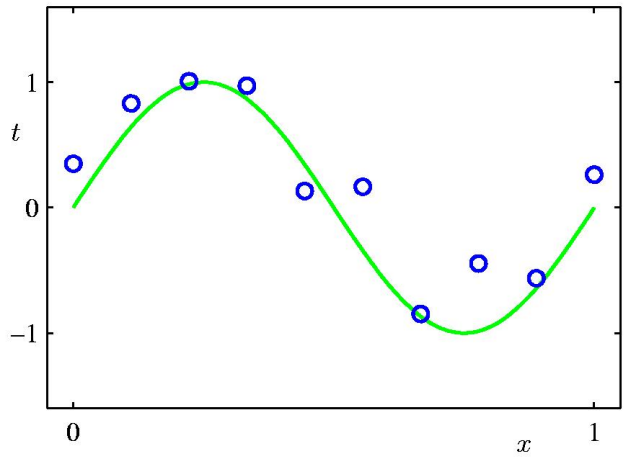
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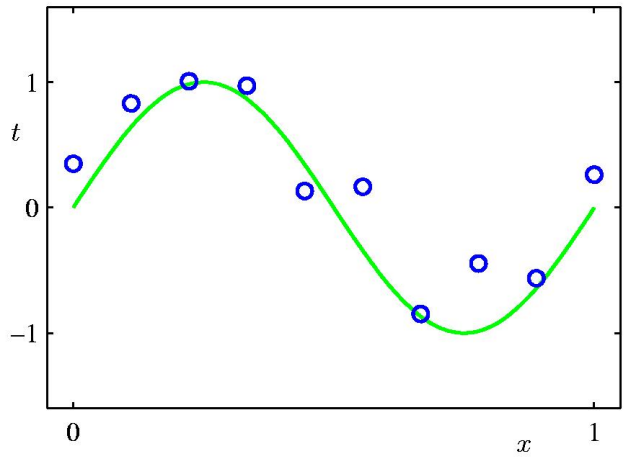
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  - Additional attributes not taken into account by data attributes, affect target values (latent variables). In the example, what else could affect house prices?
  - Model may be too simple to account for data targets

Least-Squares Regression



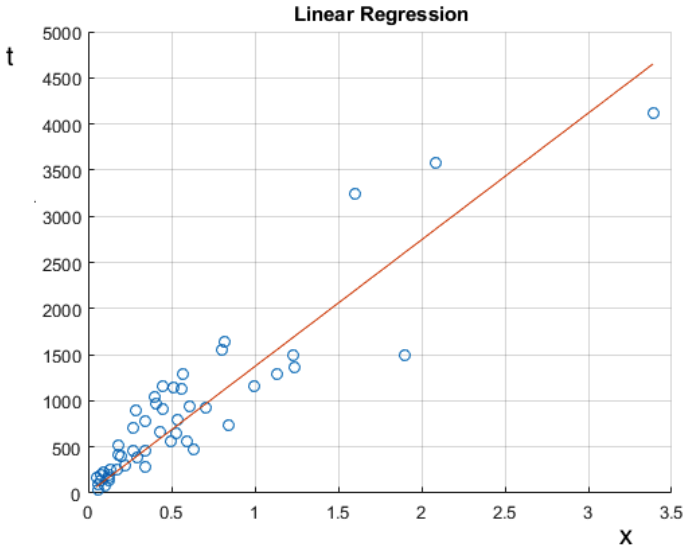
Least-Squares Regression



- Define a model

$$y(x) = \text{function}(x, \mathbf{w})$$

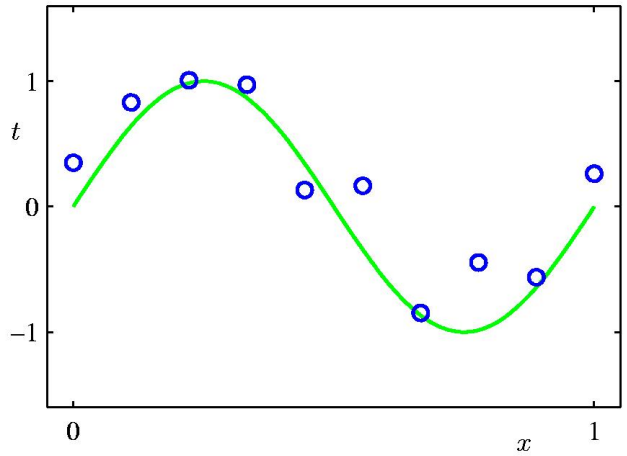
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$$\text{Linear:} \qquad y(x) = w_0 + w_1 x$$

Least-Squares Regression



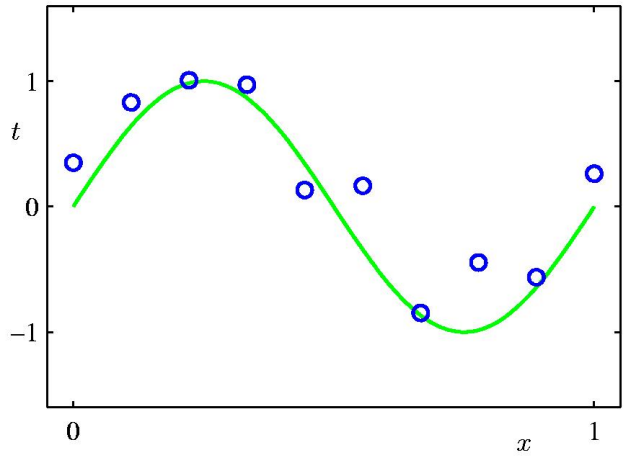
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- Standard loss/cost/objective function measures the squared error between y and the true value t

$$\ell(\mathbf{w}) = \sum_{n=1}^{N} [t^{(n)} - y(x^{(n)})]^2$$

Least-Squares Regression



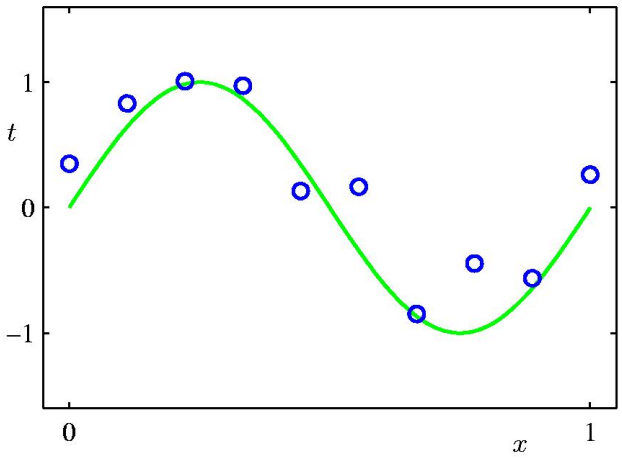
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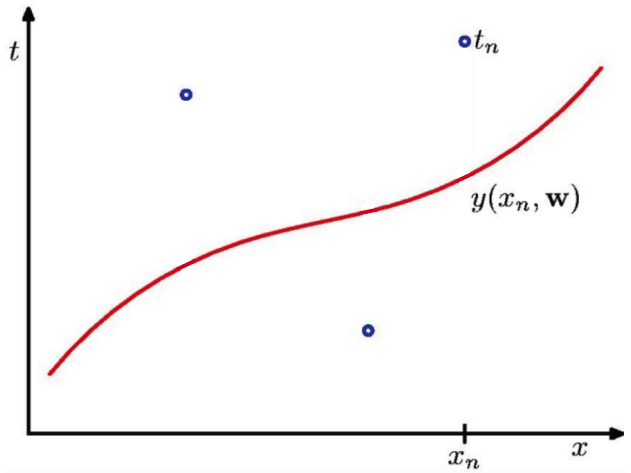
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Least-Squares Regression





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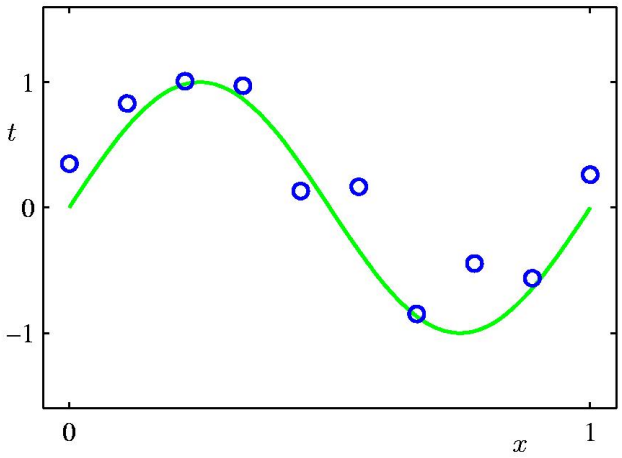
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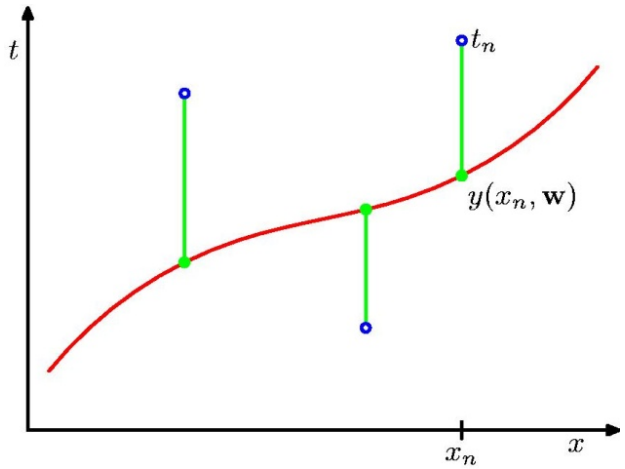
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- For a particular hypothesis ($y(x)$ defined by a choice of $\mathbf{w}$, drawn in red), what does the loss represent geometrically?

Least-Squares Regression





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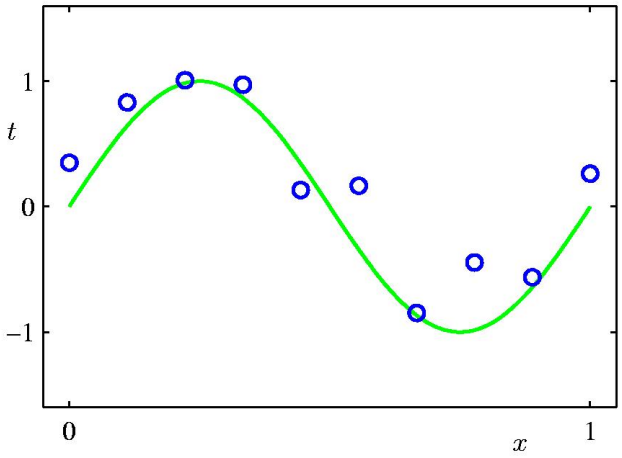
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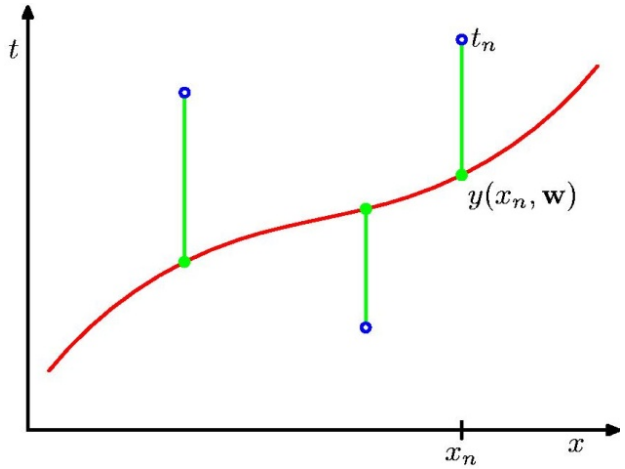
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- The loss for the red hypothesis is the **sum of the squared vertical errors** (squared lengths of green vertical lines)

Least-Squares Regression





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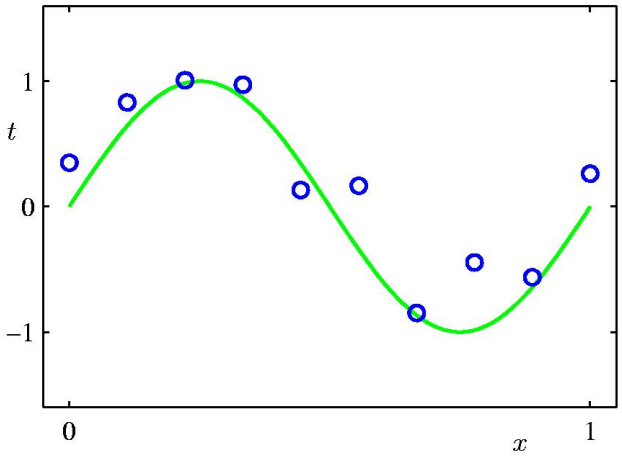
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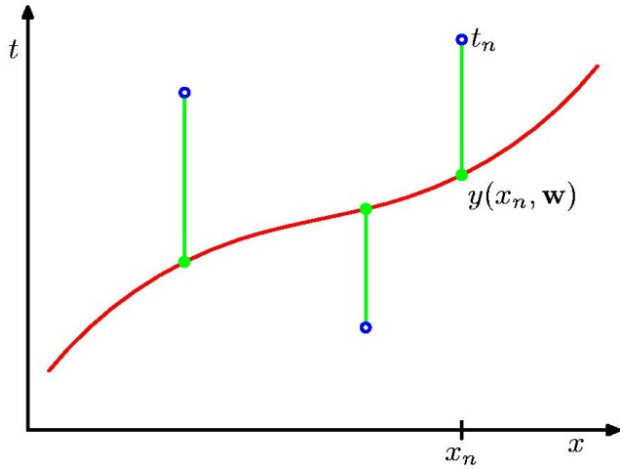
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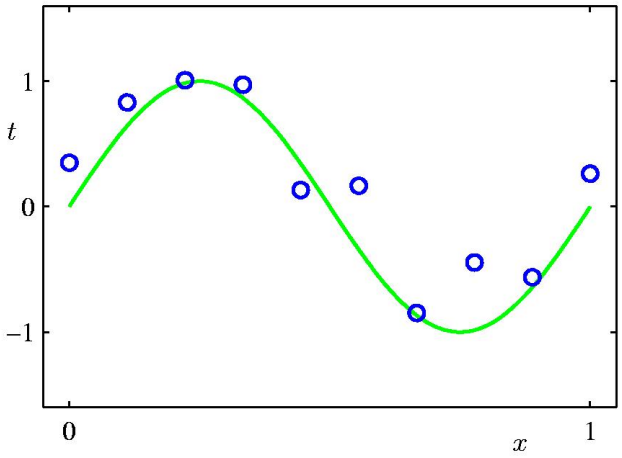
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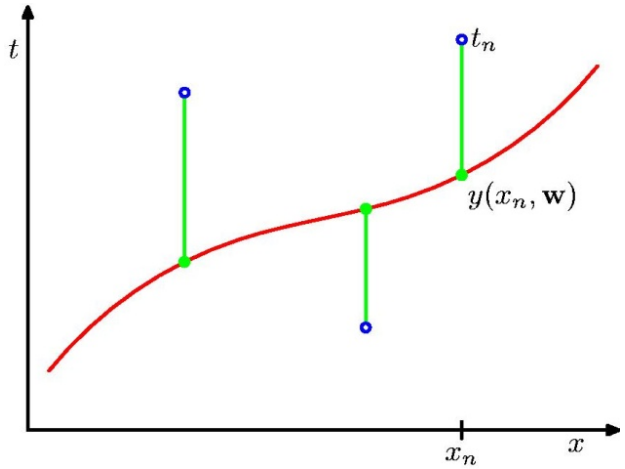
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- How do we obtain weights $\mathbf{w} = (w_0, w_1)$? Find $\mathbf{w}$ that minimizes loss $\ell(\mathbf{w})$

Least-Squares Regression





- Define a model

$$\text{Linear:} \qquad y(x) = w_0 + w_1 x$$

- Standard loss/cost/objective function measures the squared error between y and the true value t

$$\text{Linear model:} \qquad \ell(\mathbf{w}) = \sum_{n=1}^{N} [t^{(n)} - (w_0 + w_1 x^{(n)})]^2$$

- How do we obtain weights $\mathbf{w} = (w_0, w_1)$?
- For the linear model, what kind of a function is $\ell(\mathbf{w})$?